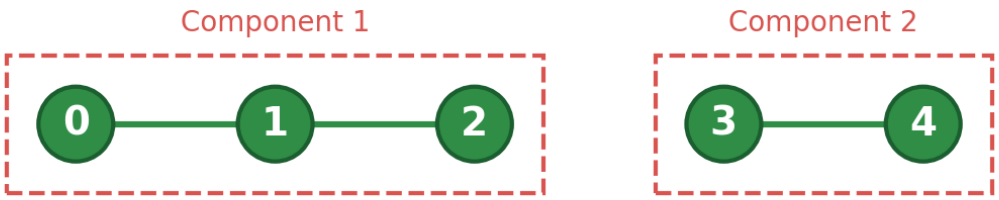
Number of Connected Components

Difficulty: **Medium** | Points: 4 | Topics: DFS, Graph, Connected Components in an Undirected Graph

Given an undirected graph with **V** vertices numbered from **0** to **V-1** and **E** edges, represented as a 2D array **edges[][]**, where each entry **edges[i] = [u, v]** denotes an edge between vertices **u** and **v**. Count the number of connected components in the graph. Two vertices belong to the same connected component if there is a path between them.

Example 1:

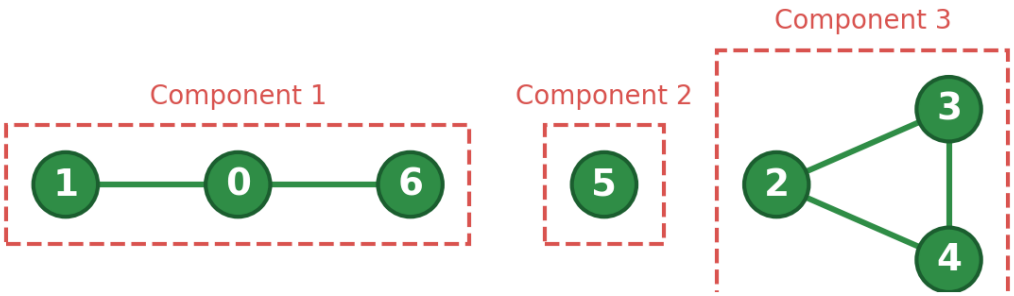
```
Input: V = 5, edges[][] = [[0, 1], [2, 1], [3, 4]]
Output: 2
```



Explanation: Vertices 0, 1 and 2 are connected to each other, and vertices 3 and 4 are connected to each other, so there are 2 connected components.

Example 2:

```
Input: V = 7, edges[][] = [[0, 1], [6, 0], [2, 4], [2, 3], [3, 4]]
Output: 3
```



Explanation: Vertices {0, 1, 6} form one component, {2, 3, 4} form another, and vertex 5 is isolated, so there are 3 connected components.

Constraints:

$1 \leq V \leq 10^4$

Expected Complexity:

Time Complexity: $O(V^2)$ Auxiliary Space: O(V)

Source: https://www.geeksforgeeks.org/problems/number-of-provinces/1